Physics 4B Spring 2026
Week 3 Discussion

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1. **Brainwarmer**.

(a) Two tuning forks are struck simultaneously. The first has a frequency of 440 Hz. An observer hears a "beat" once every 0.25 seconds. (i) Find the beat frequency. (ii) What are the possible frequencies of the second tuning fork? (iii) Write a formula that gives the time dependence of the resulting wave.

Solution:

(i) The period of the beats is $T_b = 0.25$ s, so the beat frequency is $f = \frac{1}{T_b} = 4$ Hz.

(ii) $f_b = |f_1 - f_2|$, so f_2 is either 436 Hz or 444 Hz.

(iii) Let the initial waves be described by

$$\psi_1(x,t) = A\cos\left(\frac{2\pi f_1}{v_s}x - 2\pi f_1 t\right),\ \psi_2(x,t) = A\cos\left(\frac{2\pi f_2}{v_s}x - 2\pi f_2 t\right) \tag{1}$$

The total wave is

$$\begin{aligned}\psi(x,t) &= \psi_1(x,t) + \psi_2(x,t)\\ &= 2A\cdot\cos\left(\frac{\pi(f_1+f_2)}{v_s}x - \pi(f_1+f_2)t\right)\\ &\quad\cdot\cos\left(\frac{\pi(f_1-f_2)}{v_s}x - \pi(f_1-f_2)t\right)\end{aligned} \tag{2}$$

where we used the identity $\cos a + \cos b = 2\cos((a+b)/2)\cos((a-b)/2)$. We only want the time dependence of the resulting wave, which is

$$\psi(t) = 2A\cos(\pi(f_1-f_2)t)\cos(\pi(f_1+f_2)t) \tag{3}$$

As we can see, the envelope has a frequency of $\frac{|f_1-f_2|}{2} = 2$ Hz and the carrier has a frequency of $\frac{f_1+f_2}{2} \approx 440$ Hz.

You may wonder why the beat frequency is twice the frequency of the envelope - This is because every other outburst in the modulation is inverted, so both the peaks and troughs in the envelope are perceived as beats. As a result, the beat frequency is twice the frequency of the envelope.

(b) A source of sound radiates uniformly in all directions. Along a radial line from the source locate two points separated by 2 m such that the intensity at the nearer point is 4 dB above that of the more distant point[1].

Solution: Let the two points be r_1 and r_2 from the source, with $r_1 < r_2$. The difference in intensity between the two points is

$$4 = 10 \log_{10} \left(\frac{I_1}{I_2} \right) = 10 \log_{10} \left(\frac{r_2^2}{r_1^2} \right) = 20 \log_{10} \left(\frac{r_1 + 2 \text{ m}}{r_1} \right) \tag{4}$$

Solve for r_1, we get

$$r_1 = \frac{2 \text{ m}}{10^{1/5} - 1} \approx 3.42 \text{ m} \tag{5}$$

2. **Interference**. I am at my yacht on the beach at the point $(0, 0)$, where the beach runs along the y-axis. A wall has been built at $x = 48$ m, parallel to the beach, to block water waves of wavelength $\lambda = 8$ m. A hole develops in the wall at the point directly in front of my yacht, $(48, 0)$. Where should I make a second hole, nearest to this one, so that the waves interfere destructively at my yacht and constructively at the point $(0, 10)$? Do the calculation exactly, without approximations.

Solution: Let the second hole be at $(48, y)$, and we want to find the smallest y that satisfy the conditions.

(i) Destructive interference at $(0, 0)$: The path length difference

$$|L_1 - L_2| = \sqrt{48^2 + y^2} - 48 \tag{6}$$

must be an odd half-integer multiple of the wavelength, i.e. $|L_1 - L_2| = (n + 1/2)\lambda$. The smallest y comes from taking the smallest n, so

$$\sqrt{48^2 + y^2} - 48 = \frac{\lambda}{2} = 4 \tag{7}$$

This leads to $y = \pm 20$.

(ii) Constructive interference at $(0, 10)$: For this point, the path length difference must be an integer multiple of the wavelength, i.e. $|L_1' - L_2'| = n\lambda$. We check the two candidates from the previous part:

Candidate 1 $(48, 20)$:

$$|L_1' - L_2'| = \sqrt{48^2 + 10^2} - \sqrt{48^2 + (20 - 10)^2} = 0 \tag{8}$$

This is constructive interference.

[1] Shankar 9.15

Candidate 2 $(48, -20)$:

$$|L_1' - L_2'| = \sqrt{48^2 + 10^2} - \sqrt{48^2 + (-20 - 10)^2} \approx 7.57 \tag{9}$$

This is not an integer multiple of 8, so it's not constructive interference. Therefore, the second hole should be at (48, 20).

3. **Doppler effect**. A small speaker is attached to a block of mass m on a horizontal spring. The speaker emits sound at frequency $f = 650$ Hz. The block undergoes simple harmonic motion with frequency 5.0 Hz and amplitude A. The speed of sound is $v_s = 340$ m/s. If the difference between the maximum and minimum frequencies heard by a stationary observer is 3.0 Hz, what is the amplitude A?

   Solution: We are going to assume that the spring is oscillation along a direction pointing directly at the observer. The source is moving and the observer is stationary, so we are going to use

   $$f = f_0 \frac{1}{1 \pm u/v_s} \tag{10}$$

   where u is the speed of the source relative to the medium. The minimum frequency results from the speaker moving at maximum speed away, while the maximum frequency results from the speaker moving at maximum speed towards the observer, so we need to find the maximum speed.

   For a simple harmonic oscillator, the motion is given by $x(t) = A\cos(\omega t)$, so the maximum speed is $v_{\max} = \omega A$. Therefore, we can set up the equation

   $$f_{\max} - f_{\min} = f_0 \left(\frac{1}{1 - \omega A/v_s} - \frac{1}{1 + \omega A/v_s} \right) \tag{11}$$

   Rearrange this

   $$-\omega^2 \alpha A^2 - A + v_s^2 \alpha = 0 \tag{12}$$

   where $\alpha \equiv (f_{\max} - f_{\min})/(2\omega f_0 v_s)$, with $\omega = 2\pi f_{\text{osc}} = 2\pi(5\text{ Hz})$. Only the positive solution makes physical sense, so we have

   $$A = \frac{-1 + \sqrt{1 + 4\omega^2 v_s^2 \alpha^2}}{2\omega^2 \alpha} = 0.025 \text{ m} \tag{13}$$

4. **Sound intensity**. Two people, A and B, are each equipped with a sound detector that can read both the sound intensity in its environment and the frequency of the dominant sound. A speaker that emits sound uniformly in all directions is placed in an otherwise empty quiet field. A stands stationary at a distance of 20.0 meters from the speaker, and their instrument detects a 1000 Hz sound at

an intensity of 90 dB. B is running directly away from the speaker at a constant velocity of 10 m/s. When B crosses a point that is 80 meters from the speaker, what (a) frequency and (b) intensity does their detector record? Assume the speed of sound is 345.8 m/s.

Solution:

(a) A is stationary relative to the speaker, so there's no Doppler shift. The source frequency is $f_0 = 1000$ Hz. B is receding from the speaker, so the frequency detected by B is

$$f_B = f_0 \left(1 - \frac{v_B}{v_s}\right) = 971 \text{ Hz} \tag{14}$$

(b) We know the intensity at A's location in dB

$$\beta_A = 10 \log_{10}(I_A/I_0) = 90 \tag{15}$$

Since $I = \frac{P}{4\pi r^2}$, $I_B = I_A(r_A/r_B)^2$. The intensity at B's location is then

$$\begin{aligned} \beta_B &= 10 \log_{10}(I_B/I_0) \\ &= 10 \log_{10}\left(\frac{I_A}{I_0}\frac{r_A^2}{r_B^2}\right) \\ &= \beta_A + 10 \log_{10}\left(\frac{r_A^2}{r_B^2}\right) \\ &= 90 + 20 \log_{10}(20/80) \\ &\approx 78 \end{aligned} \tag{16}$$

5. **Single-slit interference**[2]. A plane sound wave of frequency 480 Hz passes through a single slit of width a. The central diffraction maximum is observed at $0°$, and the first minima occur at $\pm 23°$. Assume the speed of sound is $v_s = 340$ m/s. (i) Find the slit width a, (ii) Find the angle for the next minimum, (iii) Why there are no more minima beyond a certain order.

Solution:

(i) For single-slit diffraction, the diffraction minima occur at $a \sin\theta = m\lambda$ with $m = 1, 2, 3, ...$

The first minima ($m = 1$) occur at $\pm 23°$, so

$$a = \frac{\lambda}{\sin(23°)} = \frac{v_s/f}{\sin(23°)} \approx 1.81 \text{ m} \tag{17}$$

[2]If time allows

(ii) The next minimum is at $m = 2$, so

$$\theta = \arcsin(m\lambda/a) = \arcsin(2v_s/f/a) \approx 51.4^\circ \tag{18}$$

(iii) Since $\sin\theta = \frac{m\lambda}{a} \leq 1$, we have $m \leq a/\lambda \approx 2.56$. This means there are only two minima on each side of the central maximum. Physically, this is due to the fact that the path length difference between the interfering waves is limited by the size of the slit.